

HDFS: command line commands

HDFS

- The content of a HDFS file can be accessed by means of
 - Command line commands
 - A basic web interface provided by Apache Hadoop
 - The HDFS content can only be browsed and its files downloaded from HDFS to the local file system
 - Uploading functionalities are not available
 - Vendor-specific web interfaces providing a full set of functionalities (upload, download, rename, delete, ...)
 - E.g., the HUE web application of Cloudera

HDFS – user folder

- Each user of the Hadoop cluster has a personal folder in the HDFS file system
 - The default folder of a user is

`/user/username`

HDFS – command line

- The **hdfs** command can be executed in a Linux shell to read/write/modify/delete the content of the distributed file system
- The parameters/arguments of hdfs command are used to specify the operation to execute

HDFS – command line

- List the content of a folder of the HDFS file system

hdfs dfs -ls *folder*

- Example

hdfs dfs -ls /user/garza

- shows the content (list of files and folders) of the /user/garza folder

HDFS – command line

- Example

hdfs dfs -ls .

- shows the content of the home of the current user
 - i.e., the content of /user/*current_username*
 - . = user home
- The mapping between the local linux user and the user of the cluster is based on
 - A Kerberos ticket if Kerberos is active
 - Otherwise the local linux user is considered

HDFS – command line

- Show the content of a file of the HDFS file system

hdfs dfs -cat *file*

- Example

hdfs dfs -cat /user/garza/document.txt

- Shows the content of the /user/garza/document.txt file stored in HDFS

HDFS – command line

- Copy a file from the local file system to the HDFS file system

hdfs dfs -put *local_file HDFS_path*

- Example

hdfs dfs -put /data/document.txt /user/garza/

- Copy the local file /data/document.txt in the folder /user/garza of HDFS

HDFS – command line

- Copy a file from the HDFS file system to the local file system
`hdfs dfs -get HDFS_path local_file`
- Example
`hdfs dfs -get /user/garza/document.txt /data/`
- Copy the HDFS file /user/garza/document.txt in the local file system folder /data/

HDFS – command line

- Delete a file from the HDFS file system
hdfs dfs -rm *HDFS_path*
- Example
hdfs dfs -rm /user/garza/document.txt
- Delete from HDFS the file
/user/garza/document.txt

HDFS – command line

- There are many other linux-like commands
 - rmdir
 - du
 - tail
 - ...
- Useful link
 - <https://hadoop.apache.org/docs/r2.7.1/hadoop-project-dist/hadoop-hdfs/HDFSCommands.html>

HD FS and Hadoop

Hadoop – command line

- The Hadoop programs are executed (submitted to the cluster) by using the hadoop command
 - It is a command line program
 - Hadoop is characterized by a set of parameters
 - E.g., the name of the jar file containing all the classes of the MapReduce application we want to execute
 - The name of the Driver class
 - The parameters/arguments of the MapReduce application

Hadoop – command line example

(1)

- The following command executes/submits a MapReduce application

```
hadoop jar MyApplication.jar  
it.polito.bigdata.hadoop.DriverMyApplication 1  
inputdatafolder/ outputdatafolder/
```

- It executes/submits the application contained in MyApplication.jar

Hadoop – command line example (2)

- The Driver Class is
 - it.polito.bigdata.hadoop.DriverMyApplication
- The application has three arguments
 - Number of reducers (args[0])
 - Input data folder (args[1])
 - Output data folder (args[2])